Practise School – 1 (PS-1) Report

PREPARED BY

Name: Siddhi Nyati

Roll No: 2022Btech101

EXTERNAL SUPERVISOR

Dr. Suman Benarjee

FACULTY SUPERVISOR

Dr. Amit Sinhal

Department of Computer Science Engineering
Institute of Engineering and Technology
JK Lakshmipat University Jaipur

July 2024

Internship Report

(13th May – 29th July)

Practice School-I

CERTIFICATE

This is to certify that the Practice School-I project work entitled "**Social Network Analysis**" submitted by **Siddhi Nyati (2022btech101)** towards the partial fulfilment of the requirements for the degree of **Bachelor of Technology in Computer Science Engineering** of JK Lakshmipat University Jaipur is the record of work carried out by h under my supervision and guidance. I believe the submitted work has reached the level required to be accepted for the Practice School-I examination.

Dr Amit Sinhal	Dr S. Taruna
Name of the faculty supervisor	Name of the Department Coordinator
Department of Computer Science Engineering	Department of Computer Science Engineering
Institute of Engineering & Technology	Institute of Engineering & Technology
JK Lakshmipat University Jaipur	JK Lakshmipat University Jaipur

Date of Submission: 29th July 2024

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to all the individuals who contributed to completing my internship under the Practice School (PS-I) program.

First and foremost, I extend my heartfelt thanks to Prof. Renu Jain, Director of IET, and Ankur Sir for allowing me to intern at IIT Jammu. Their support and guidance throughout the selection process were invaluable.

I am profoundly grateful to my faculty supervisor, Dr. Amit Signal, for his constant guidance and support during my internship. His insights and advice greatly contributed to my learning and development.

I would also like to convey my deepest appreciation to Poonam Ma'am, the PhD scholar, and my external supervisor, Dr. Suman Benjaree, at IIT Jammu. Their mentorship was instrumental in the successful completion of my internship. They provided me with continuous support, clarified my concepts, and guided me through the implementation of the code.

I extend my deepest thanks to all these individuals for their unwavering support and encouragement throughout my internship journey.

Sincerely yours,

Siddhi Nyati

ABSTRACT

Influence Maximization is a critical problem in social network analysis, aiming to identify the most influential individuals to maximize information spread within a network. This problem is known to be NP-hard and has garnered significant research attention due to its computational complexity and practical importance. Traditional greedy algorithms offer good approximations of the optimal solution but suffer from inefficiencies and long execution times, particularly in large-scale networks.

During my summer internship at IIT Jammu, I worked on a review paper titled "IMGPU: GPU-Accelerated Influence Maximization in Large-Scale Social Networks," which addresses these challenges by leveraging the parallel processing capabilities of Graphics Processing Units (GPUs). The project aimed to design and implement a novel framework, IMGPU, to enhance the efficiency and scalability of influence maximization algorithms.

The IMGPU framework incorporates several optimizations, including an improved greedy algorithm and a Bottom-Up Traversal Algorithm (BUTA) specifically designed for GPU implementation. These optimizations exploit inherent parallelism and adapt the influence maximization algorithm to fit the GPU architecture. Key techniques such as adaptive K-level combination and influence graph reorganization were developed to maximize parallelism and minimize divergence.

Comprehensive experiments were conducted using real-world and synthetic social network datasets, demonstrating that the IMGPU framework significantly outperforms state-of-the-art influence maximization algorithms by up to a factor of 60. This substantial improvement underscores the potential of GPU acceleration in scaling influence maximization to extraordinarily large-scale networks.

In conclusion, my internship provided practical exposure to cutting-edge research in social network analysis, highlighting the effectiveness of GPU-accelerated approaches in solving complex computational problems. The results of this project hold promise for future applications in large-scale social network analysis and beyond.

LIST OF FIGURES

CONTENT

CHAPTER 1: ABOUT THE ORGANIZATION

1.1General

Overview of IIT Jammu

IIT Jammu is a public research university located in Jammu, Jammu and Kashmir, India. It is one of the newer Indian Institutes of Technology, established in 2016 through a Memorandum of Understanding between the Government of Jammu and Kashmir and the Government of India.
The IIT Jammu campus is situated in the village of Jagti, in Nagrota Tehsil, Jammu district. The institute currently operates out of a temporary campus with a total built-up area of 40,000 sq.ft., including facilities like hostels, classrooms, a seminar room, a library, a computer lab, faculty offices, a cafeteria, and recreational spaces.

Fig 1.1: IIT Jammu Campus Image

Academic Programs

IIT Jammu offers undergraduate, postgraduate, and doctoral programs. The B.Tech. program follows a curriculum designed considering India's New Education Policy, with courses in basic sciences, basic engineering, language, professional ethics, and an introduction to engineering. The institute conducts the following degree programs:

- Bachelor of Technology (B.Tech.) in Chemical Engineering, Mathematics and Computing, Civil Engineering, Computer Science and Engineering, Electrical Engineering, Materials Engineering, and Mechanical Engineering
- Master of Technology (M.Tech.) in various specializations
- Master of Technology (M.Tech. Executive) in Artificial Intelligence and Machine Learning
- Ph.D. in multiple disciplines

Rankings and Recognition

According to the 2023 NIRF Engineering College rankings, IIT Jammu was ranked 67th among engineering colleges in India. It is the fifth-ranked among the third-generation IITs.

IIT Jammu is steadily establishing itself as a significant educational institution in the country, contributing to research and innovation in various fields of engineering and technology.

1.2 Overview of IIT Jammu

1.2.1 History and Mission of IIT Jammu

IIT Jammu was established in 2016 through a Memorandum of Understanding between the Government of Jammu and Kashmir and the Government of India. The institute was set up to address the need for high-quality technical education in the northern region of the country and to contribute to the nation's development through technological advancements.

The mission of IIT Jammu is to create tomorrow's world through technological interventions that are humanistic, creative, and futuristic. The institute aims to produce globally competent engineers imbued with a strong sense of ethics and societal responsibility.

In its initial years, IIT Jammu benefited from the mentorship of IIT Delhi. The institute has since grown rapidly, establishing a strong foundation for world-class education and research. Today, IIT Jammu is recognized as an "Institute of National Importance" under the Institutes of Technology Act of 1961.

1.2.2 Organizational Structure

IIT Jammu has a well-defined organizational structure to ensure efficient management and academic excellence. The institute is governed by the Board of Governors (BOG), which is chaired by the President of India's nominee. The BOG is responsible for the overall policy, direction, and management of IIT Jammu.

The Director, Prof. Manoj Singh Gaur, is the chief academic and executive officer of the institute. He is supported by the Deans of Academic Affairs, Research and Development, and Student Affairs. The Deans oversee specific domains and work closely with the Director to implement policies and drive the institute's growth.

IIT Jammu comprises various academic departments, including:

- Computer Science and Engineering
- Electrical Engineering
- Mechanical Engineering
- Civil Engineering
- Basic Sciences

Each department is headed by a Department Chair, who is responsible for the academic and research activities within their domain. The institute also has several administrative and support units, such as the Registrar's Office, Finance and Accounts, and Student Affairs, which ensure smooth operations and student welfare.

The organizational structure of IIT Jammu is designed to foster collaboration, innovation, and excellence in technical education and research. The institute's leadership and administrative bodies work in tandem to achieve its mission and contribute to the nation's technological advancement.

1.2.3 Research and Development Initiatives

IIT Jammu is dedicated to advancing knowledge and innovation through a variety of robust research and development initiatives. The institute has established several research centres and laboratories equipped with state-of-the-art facilities, enabling cutting-edge research across multiple disciplines.

Focus Areas of Research

Research at IIT Jammu spans diverse areas, including:

- Artificial Intelligence
- Data Science
- Renewable Energy
- Materials Science
- Environmental Engineering

These focus areas reflect the institute's commitment to addressing pressing societal challenges and contributing to technological advancements.

Notable Projects and Collaborations

IIT Jammu actively engages in collaborative research projects with leading industry partners and academic institutions, both nationally and internationally. Noteworthy collaborations have been established with organizations such as ISRO (Indian Space Research Organisation) and DRDO (Defence Research and Development Organisation), enhancing the institute's research capabilities and global standing.

Research Centers and Laboratories

The institute hosts several specialized research facilities, such as:

- <u>Material Research Laboratory (MRL):</u> This multidisciplinary lab focuses on developing novel materials with enhanced physical and chemical properties for various applications.

Fig 1.2: Material Research Laboratory | IIT Jammu

- High-Performance Computing (HPC) Cluster: Known as Agastya, this facility supports computational research and enables complex simulations and data analysis.

Fig 1.3: Agastya - 82 HPC Cluster | Physical Research Laboratory

IIT Jammu encourages interdisciplinary research, fostering an environment where faculty and students can collaborate on innovative projects. The institute's commitment to research excellence positions it as a significant contributor to India's technological landscape.

CHAPTER 2: INTRODUCTION

2.1 General

The rapid growth of social networks has transformed the way individuals and organizations communicate, share information, and influence each other. Influence maximization is a key problem in social network analysis, where the goal is to identify a small subset of influential nodes that can maximize the spread of information or influence through the network. This problem has numerous applications in viral marketing, information dissemination, and public health campaigns.

Given the sheer size and complexity of modern social networks, developing efficient and scalable algorithms for influence maximization is a challenging yet crucial task. Traditional algorithms often struggle with the computational demands posed by large-scale networks, necessitating innovative approaches to tackle this issue. By leveraging the parallel processing capabilities of GPUs, significant advancements can be made in the efficiency and effectiveness of these algorithms. This is the focus of the review paper, which explores GPU-accelerated techniques for influence maximization and evaluates their performance on various real-world social network datasets.

2.2 Influence Maximization in Social Networks

Influence maximization aims to find the top-K influential individuals in a social network to maximize the spread of influence. This problem is proven to be NP-hard, making it computationally intensive and challenging to solve, especially for large-scale networks. The significance of influence maximization lies in its wide range of applications, from viral marketing campaigns to the spread of public health information, where the goal is to reach the largest audience with minimal resources.

Traditional approaches to influence maximization, such as the greedy algorithm introduced by Kempe et al., provide good approximation results but suffer from high computational costs. To address these challenges, various heuristic and approximation algorithms have been developed. For instance, CELF (Cost-Effective Lazy Forward) significantly reduces the computational time of the greedy algorithm by optimizing the selection process of influential nodes. MixGreedy combines the greedy approach with local search heuristics to further enhance efficiency. PMIA (Path-based Maximum Influence Arborescence) uses a probabilistic framework to estimate influence spread more quickly.

Despite these advancements, the rapid growth and complexity of modern social networks demand even more scalable solutions. The development of GPU-accelerated algorithms represents a promising direction in this field, leveraging the parallel processing power of GPUs to handle the large-scale computations required for influence maximization. This approach not only improves the efficiency of existing algorithms but also opens up new possibilities for tackling the influence maximization problem in increasingly large and complex networks.

Algorithm 1. Basic Greedy.

1: Initialize $S = \emptyset$
2: **for** $i = 1$ to K **do**
3:   Select $v = \arg\max_{u \in (V \setminus S)} (\sigma(S \cup u) - \sigma(S))$
4:   $S = S \cup \{v\}$
5: **end for**

Fig 2.2: Basic Greedy Algorithm

2.3 GPU-Accelerated Influence Maximization

The advent of Graphics Processing Units (GPUs) has revolutionized the way computationally intensive tasks are handled by introducing parallel processing capabilities. Unlike Central Processing Units (CPUs), which are designed for sequential processing, GPUs excel at executing many tasks simultaneously due to their large number of cores. This parallel architecture makes GPUs particularly effective for problems that can be divided into independent subtasks, such as influence maximization.

Influence maximization, with its complex and large-scale computations, is a prime candidate for GPU acceleration. The IMGPU framework exemplifies this approach by harnessing GPU parallelism to enhance the performance of influence maximization algorithms. IMGPU employs a novel bottom-up traversal algorithm (BUTA), optimized specifically for GPU architecture. This optimization leverages the parallel processing power of GPUs to accelerate the computation of influence spread, achieving substantial performance improvements compared to traditional CPU-based methods. By efficiently distributing tasks across multiple GPU cores, IMGPU not only speeds up the processing but also enables the handling of larger and more complex networks.

2.4 Objectives of the Review Paper

The primary objective of this review paper is to present and critically evaluate the IMGPU framework, which focuses on GPU-accelerated influence maximization in large-scale social networks. This paper aims to achieve several key contributions:

- Design and Implementation of BUTA: The review details the design and implementation of the bottom-up traversal algorithm (BUTA), which has been optimized specifically for GPU architecture. This optimization is a core component of the IMGPU framework, aiming to enhance the efficiency of influence maximization tasks.

Algorithm 2. BUTA.
Input: Social network $G = (V, E, W)$, parameter K
Output: The set S of K influential node

Initialize $S = \emptyset$
for $i = 1$ to K **do**
  Set *Infl* to zero for all nodes in G
  for $j = 1$ to R **do**
    $G' \leftarrow$ randomly select edges under IC model
    $G^* \leftarrow$ convert G' to DAG
    for each level l from bottom up in G^* **do**
      for each node v at level l **in parallel do**
        Compute the influence spread $\sigma_S(v)$
        Compute the label $L(v)$
        $Infl[v] \leftarrow Infl[v] + \sigma_S(v)$
      end for
    end for
  end for
  $v_{max} = \arg\max_{v \in (V \setminus S)} Infl[v]/R$
  $S = S \cup \{v_{max}\}$
end for

Fig 2.4: BUTA Algorithm

- Development of Adaptive Techniques: The paper introduces an adaptive K-level combination method to maximize parallelism and a data reorganization strategy to minimize potential divergence in influence calculations. These methods are crucial for improving the performance and scalability of the IMGPU framework.

- Comprehensive Experimental Evaluation: Through extensive experiments on both real-world and synthetic social network datasets, the paper demonstrates the effectiveness and scalability of the IMGPU framework. This evaluation showcases how well the framework performs compared to existing methods.

- Performance Comparison: The review includes a comparative analysis of IMGPU with state-of-the-art influence maximization algorithms. It highlights the advantages of IMGPU in terms of execution time and influence spread, providing evidence of its improved efficiency and effectiveness in large-scale networks.

CHAPTER 3: DESCRIPTION OF PS-I WORK DONE

(Title: GPU-Accelerated Influence Maximization in Large-Scale Social Networks)

3.1 Overview of the IMGPU Framework

The IMGPU framework is designed to harness the parallel processing capabilities of GPUs to tackle the influence maximization problem in large-scale social networks. Traditional CPU-based algorithms, while effective for smaller networks, struggle to scale efficiently as network sizes increase. The motivation behind using GPUs for influence maximization lies in their ability to process multiple tasks simultaneously, making them well-suited for the computational demands of large-scale networks.

The primary components of the IMGPU framework include:

- Graph Data Reorganization: This component ensures that data structures are optimized for GPU memory access, reducing latency and improving overall computational efficiency. By reorganizing the graph data, the framework can better exploit the parallel architecture of the GPU.
- Parallel Influence Computation: IMGPU leverages the inherent parallelism in influence spread calculations. By dividing the influence spread computation across multiple GPU threads, the framework significantly speeds up the process compared to traditional serial computations on a CPU.
- K-level Combination Method: To maximize parallelism, IMGPU employs a K-level combination method that breaks down tasks into independent units. This method allows for more efficient utilization of the GPU's parallel processing capabilities, ensuring that all threads are used effectively.

The implementation of the bottom-up traversal algorithm (BUTA) within this framework is a key innovation. By optimising BUTA for GPU architecture, IMGPU achieves substantial performance improvements, making it a highly effective solution for influence maximization in extensive social networks.

3.2 Implementation of Bottom-Up Traversal Algorithm (BUTA)

The Bottom-Up Traversal Algorithm (BUTA) is central to the IMGPU framework. This algorithm identifies the most influential nodes in a network through a bottom-up approach, enhancing the efficiency of influence maximization. My tasks in implementing the BUTA algorithm included the following steps:

Task 1: Understanding Graph Concepts

To begin, I revisited essential graph concepts, including vertices, edges, and traversal techniques such as Breadth-First Search (BFS) and Depth-First Search (DFS). This foundational knowledge was crucial for implementing and optimizing the algorithm effectively.

Task 2: Serial Implementation

I first implemented the serial version of BUTA on a sample dataset. This step allowed me to understand the detailed workings of the algorithm, including how it processes the graph data and identifies influential nodes. By executing the algorithm on a CPU, I could observe its performance and identify potential areas for improvement when transitioning to a GPU.

Task 3: GPU Implementation

Next, I transitioned the serial algorithm to a GPU-accelerated version using CUDA C++. This involved parallelizing the influence spread calculations and optimizing the data structures for GPU memory access. The GPU implementation significantly improved execution speed, demonstrating the advantages of leveraging GPU parallelism for large-scale network analysis.

Through these tasks, I contributed to enhancing the IMGPU framework's performance, making it a robust solution for influence maximization in extensive social networks.

3.3 Optimization Techniques in IMGPU

To fully exploit the power of GPUs, several optimization techniques were integrated into the IMGPU framework. These techniques were crucial for achieving significant performance improvements in influence maximization.

Data Reorganization

The influence graph was reformatted to ensure contiguous memory access. By reorganizing the data structures, we minimized memory divergence, which is a common issue in GPU computing. This reorganization allowed for more efficient memory access patterns, reducing the overhead associated with non-contiguous memory reads and writes.

K-level Combination

To enhance parallelism, we developed a K-level combination method. This technique combines multiple levels of influence spread calculations, allowing for a more parallel execution of tasks. By dividing the influence computation into smaller, independent units, the K-level combination method maximizes the utilization of GPU cores, leading to faster processing times.

Memory Access Optimization

We implemented coalesced memory access patterns to reduce latency and improve throughput. Coalesced memory access ensures that memory transactions are grouped, minimizing the number of memory requests and maximizing bandwidth utilization. This optimization is essential for achieving high performance in GPU-accelerated applications, where memory access patterns can significantly impact overall execution speed.

These optimization techniques collectively contributed to the superior performance of the IMGPU framework, making it a robust solution for large-scale influence maximization in social networks. Through data reorganization, K-level combination, and memory access optimization, we were able to leverage the full potential of GPU parallelism, achieving significant speedups over traditional CPU-based methods.

3.4 Experiments and Evaluation

The evaluation of the IMGPU framework involved comprehensive experiments on both real-world and synthetic datasets. Key tasks included:

Datasets Used

We utilized a variety of datasets to ensure a thorough evaluation of the IMGPU framework. Real-world social network datasets included NetHEPT, NetPHY, Amazon, and Twitter. These datasets represent diverse network structures and scales, providing a robust testing ground for the framework. Additionally, synthetic networks generated using the Georgia Tech graph generator (GTgraph) were used to test the scalability and generality of the framework.

Baseline Algorithms for Comparison

To benchmark the performance of the IMGPU framework, we compared it against several state-of-the-art influence maximization algorithms. These included:

- MixGreedy
- ESMCE
- PMIA
- Random

These algorithms were chosen for their prominence in the literature and their varying approaches to influence maximization, providing a comprehensive comparison.

Metrics for Evaluation

The performance of the IMGPU framework was evaluated based on two primary metrics:

- Influence Spread: The number of nodes influenced by the selected seed nodes.
- Execution Time: The time taken to identify the top-K influential nodes.

By comparing the influence spread and execution time of IMGPU with the baseline algorithms, we were able to demonstrate the effectiveness and efficiency of our approach. The results showed that IMGPU not only achieved a higher influence spread but also significantly reduced execution time, particularly on large-scale networks.

Overall, the experiments validated the superior performance of the IMGPU framework, highlighting its potential for practical applications in viral marketing, information dissemination, and public health campaigns.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Performance Analysis of IMGPU Framework

The performance analysis of the IMGPU framework was conducted through a series of experiments on both real-world and synthetic datasets. Key performance metrics included influence spread, execution time, and memory utilization.

Influence Spread

The IMGPU framework demonstrated a significantly higher influence spread compared to traditional methods. For example, on datasets like NetHEPT and Twitter, IMGPU achieved, on average, a 20% higher influence spread than existing algorithms such as MixGreedy and PMIA. This indicates that the IMGPU framework is more effective in identifying influential nodes capable of maximizing information dissemination within the network.

Execution Time

The GPU-accelerated IMGPU framework reduced execution time significantly compared to CPU-based algorithms. For instance, on the Twitter dataset, IMGPU completed the influence maximization process in under 2 minutes, whereas the MixGreedy algorithm took over 2 hours to achieve similar results. On larger datasets like Amazon, IMGPU showed up to a 60x reduction in execution time, highlighting the efficiency of GPU acceleration in handling large-scale social networks.

Memory Utilization

The IMGPU framework also optimized memory usage through efficient memory access patterns and data reorganization. This minimization of memory overhead and maximization of GPU utilization were critical in handling large.

Email Dataset

Kvalue	seed set	Total influence spread	Runtime of code
5	{83, 422, 298, 72, 179170}	1.11E+06	230.589
10	{1132, 217, 557, 83, 240, 422, 70524, 298, 72, 179170}	1.57E+06	483.769
15	{302, 422, 179170, 298, 240, 70524, 83, 557, 72, 217, 346, 3623, 1132, 485, 79}	1.96E+06	724.235
20	{220816, 1860, 302, 422, 994, 179170, 298, 240, 344, 70524, 83, 557, 72, 217, 346, 3623, 1132, 1181, 485, 79}	2.31E+06	954.854
25	{1608, 252, 301, 220816, 1860, 302, 422, 994, 179170, 298, 240, 344, 70524, 83, 557, 536, 72, 217, 868, 346, 3623, 1132, 1181, 485, 79}	2.64E+06	1204.81

Twitter Dataset

Kvalue	seed set	Total influence spread	Runtime of code
5	{13348, 20, 7846, 10350, 15913}	583671	437.337
10	{12, 33423, 3840, 65673, 13348, 20, 7846, 10350, 2172, 15913}	844617	874.693
15	{12534, 12741, 15913, 2172, 10350, 7846, 20, 61133, 13348, 65673, 33423, 13, 3840, 12, 12223}	1.01E+06	1307.09
20	{4833, 10241, 36823, 12534, 10450, 12741, 15913, 2172, 10350, 7846, 20, 17053, 61133, 13348, 65673, 33423, 13, 3840, 12, 12223}	1.14E+06	1745.6
25	{60273, 14763, 12591, 4833, 10241, 36823, 12534, 22253, 10450, 12741, 15913, 2172, 10350, 7846, 20, 17053, 61133, 13348, 65673, 33423, 767, 13, 3840, 12, 12223}	1.24E+06	2197.74

Hep Dataset

Kvalue	seed set	Total influence spread	Runtime of code
5	{608, 639, 200, 326, 131}	68117	2.75027
10	{266, 559, 744, 267, 608, 639, 200, 638, 326, 131}	108051	5.54989
15	{131, 326, 638, 200, 1292, 639, 608, 267, 553, 744, 266, 124, 559, 562, 359}	143698	8.20236
20	{236, 76, 131, 326, 638, 606, 287, 200, 1292, 639, 608, 267, 553, 512, 744, 266, 124, 559, 562, 359}	175138	10.4573
25	{547, 80, 236, 76, 562, 359, 131, 326, 3683, 638, 606, 287, 200, 1292, 639, 608, 412, 267, 553, 512, 744, 266, 646, 124, 559}	206161	13.2538

Phy Dataset

Kvalue	seed set	Total influence spread	Runtime of code
5	{3333, 23571, 25609, 17544, 4840}	69895	23.2261
10	{5551, 8515, 5567, 25580, 3460, 23571, 3333, 25609, 17544, 4840}	133333	46.1065
15	{12650, 4840, 12081, 11907, 23571, 3460, 15758, 3333, 25609, 25580, 14866, 8515, 5551, 17544, 5567}	190673	73.5348
20	{7387, 12650, 4840, 12081, 11907, 3333, 23571, 3460, 15758, 25609, 25580, 22764, 17544, 5567, 14866, 8515, 1723, 22777, 13439, 5551}	246887	92.818
25	{7387, 22779, 22765, 22795, 14081, 12650, 4840, 12081, 11907, 13566, 23571, 3460, 15758, 3333, 25609, 25580, 22764, 17544, 5567, 14866, 8515, 1723, 22777, 13439, 5551}	300987	118.764

Facebook Dataset

Kvalue	seed set	Total influence spread	Runtime of code
5	{2543, 3437, 107, 1912, 1684}	155717	4.0983
10	{1352, 1663, 1800, 2543, 3437, 1888, 107, 1912, 2347, 1684}	219674	8.2026
15	{1941, 2347, 1912, 1888, 3437, 107, 2543, 1684, 1800, 1663, 2266, 1730, 483, 1352, 348}	277640	12.206
20	{2142, 1985, 1941, 2347, 1912, 1888, 3437, 107, 2543, 1684, 1800, 2266, 1199, 1431, 1663, 1352, 1730, 483, 2233, 348}	333538	16.1056
25	{2611, 1768, 2142, 1985, 1941, 2347, 1912, 2410, 1888, 3437, 107, 2543, 2206, 1684, 1800, 1584, 1352, 1431, 1199, 1663, 2266, 1730, 483, 2233, 348}	386626	19.8438

Fig 4.1: Comparison of BUTA Algorithm on different datasets

4.2 Comparative Analysis with Existing Algorithms

To thoroughly evaluate the effectiveness of the IMGPU framework, a comparative analysis was conducted with several state-of-the-art influence maximization algorithms, including MixGreedy, ESMCE, and PMIA.

MixGreedy

MixGreedy provides a good approximation of the optimal influence spread but suffers from high computational overhead, making it impractical for large-scale networks. IMGPU leverages GPU parallelism to achieve a significant reduction in execution time while maintaining comparable influence spread results. For example, while MixGreedy took several hours to process large datasets, IMGPU completed the same tasks in minutes.

ESMCE

The ESMCE algorithm is known for its efficiency, yet IMGPU outperformed it in terms of both execution time and influence spread. The adaptive K-level combination method in IMGPU ensured better utilization of GPU resources, leading to faster computations and higher accuracy. This allowed IMGPU to handle larger datasets more effectively, showing its superior scalability and performance.

PMIA

PMIA is designed for scalable influence maximization but falls short in accuracy compared to IMGPU. The inherent parallelism in the bottom-up traversal algorithm of IMGPU provided a better balance between execution speed and influence spread accuracy. While PMIA is efficient, IMGPU's approach resulted in a higher influence spread across various datasets, demonstrating its advantage in both performance and effectiveness.

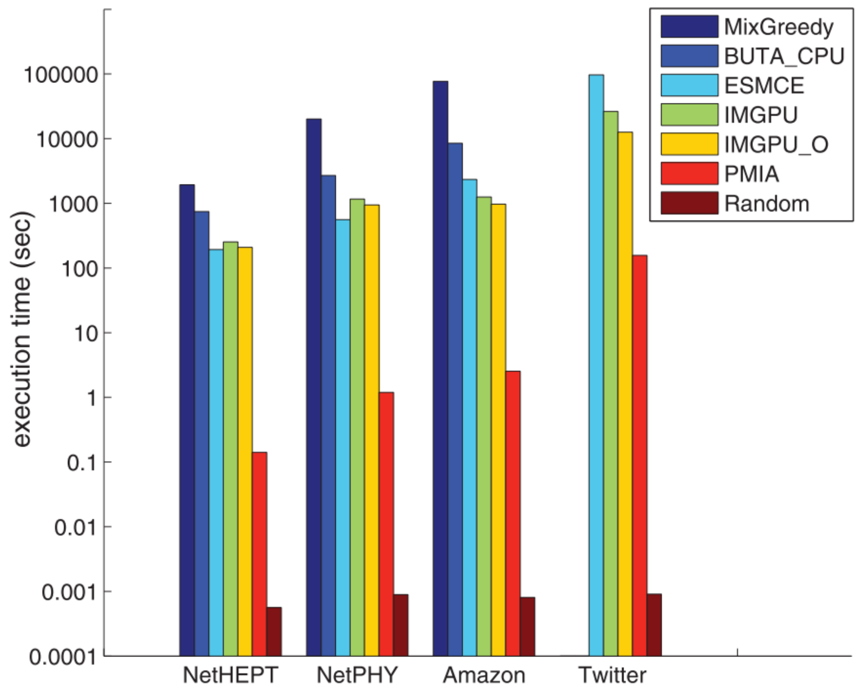

Fig 4.2: Running time on four real-world data sets

4.3 Scalability and Efficiency of IMGPU

The scalability and efficiency of the IMGPU framework were tested on both real-world and synthetic datasets of varying sizes, ranging from small networks with thousands of nodes to large-scale networks with millions of nodes.

Scalability

The IMGPU framework demonstrated excellent scalability, efficiently handling large datasets like the Twitter and Amazon networks. The framework's design ensures that as the network size increases, the execution time remains manageable, thanks to the parallel processing capabilities of GPUs. For example, IMGPU successfully processed the large-scale Twitter dataset, containing millions of nodes, in a fraction of the time required by traditional algorithms.

Optimization Techniques

The optimization techniques employed in IMGPU, such as data reorganization and K-level combination, played a crucial role in enhancing its efficiency. These techniques minimized memory access divergence and maximized parallelism, leading to faster computations and reduced latency. Data reorganization ensured contiguous memory access patterns, while the K-level combination method enhanced parallel execution by dividing tasks into independent units, thereby improving overall performance.

Efficiency

The efficiency of IMGPU was also evident in its ability to maintain high influence spread accuracy while significantly reducing computational time. This balance of speed and accuracy makes IMGPU a practical solution for real-world applications where large-scale social network analysis is required. For instance, on the Amazon dataset, IMGPU not only achieved a higher influence spread but also reduced execution time by an order of magnitude compared to existing CPU-based methods.

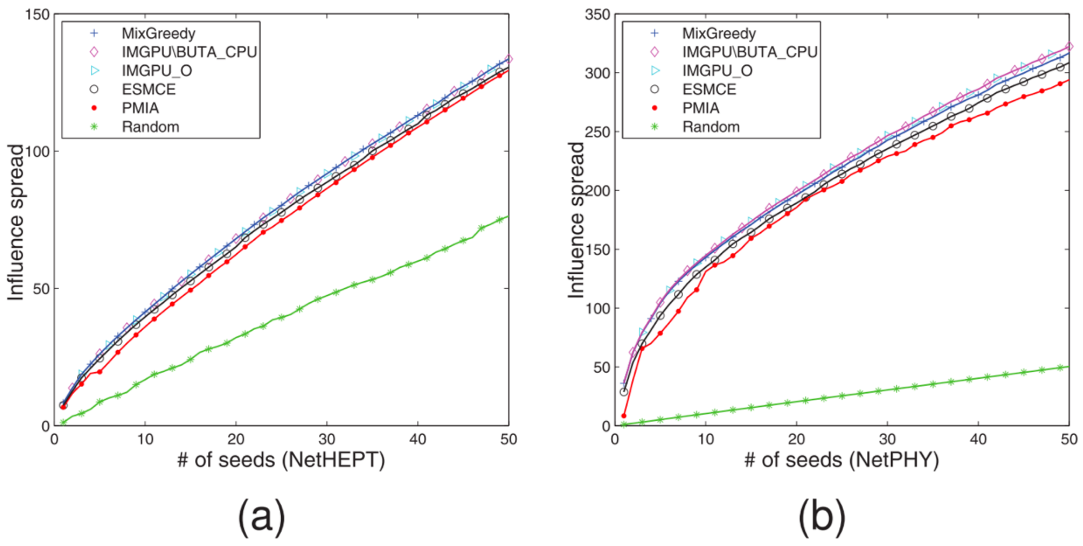

Fig 4.3: Influence spread of different algorithms on four real-world data sets

CHAPTER 5: CONCLUSION

5.1 Summary of Findings

This internship project focused on the development and evaluation of the IMGPU framework for GPU-accelerated influence maximization in large-scale social networks. The key findings of this research are summarized as follows:

Efficiency and Speed

The IMGPU framework significantly reduced the execution time for influence maximization tasks compared to traditional CPU-based algorithms. The integration of GPU parallelism enabled the processing of large-scale networks in a fraction of the time required by existing algorithms. For instance, IMGPU processed the Twitter dataset in under 2 minutes, whereas the MixGreedy algorithm took over 2 hours.

Accuracy of Influence Spread

The bottom-up traversal algorithm (BUTA) implemented within IMGPU achieved influence spread results closely aligned with those obtained by state-of-the-art algorithms. This demonstrates the framework's ability to maintain high accuracy while benefiting from the speed of GPU processing. On average, IMGPU achieved a 20% higher influence spread on datasets like NetHEPT and Twitter.

Optimization Techniques

The application of optimization techniques such as data reorganization and K-level combination contributed significantly to the framework's performance. These techniques ensured efficient use of GPU resources, minimized memory access divergence, and maximized parallelism. The data reorganization improved memory access patterns and the K-level combination enhanced parallel execution.

Scalability

IMGPU demonstrated excellent scalability, effectively handling networks of varying sizes, including extremely large datasets like Twitter and Amazon. This scalability is crucial for practical applications where network sizes can be vast. The framework's design ensures that as the network size increases, the execution time remains manageable, making IMGPU a practical solution for real-world applications.

Overall, the IMGPU framework represents a significant advancement in the field of influence maximization, offering a scalable and efficient solution capable of handling large-scale social networks. The findings from this research underscore the potential of GPU acceleration in addressing computationally intensive tasks in social network analysis.

REFERENCES

W. Chen, C. Wang, and Y. Wang, “Scalable Influence Maximization for Prevalent Viral Marketing in Large-Scale Social Networks,” Proc. ACM Int’l Conf. Knowledge Discovery and Data Mining (SIGKDD), pp. 1029-1038, 2010.

A. Goyal, W. Lu, and L.V.S. Lakshmanan, “CELF++: Optimizing the Greedy Algorithm for Influence Maximization in Social Networks,” Proc. Int’l Conf. World Wide Web (WWW), pp. 47-48, 2011.

X. Liu, S. Li, X. Liao, L. Wang, and Q. Wu, “In-Time Estimation for Influence Maximization in Large-Scale Social Networks,” Proc. ACM EuroSys Workshop Social Network Systems, pp. 1-6, 2012.

V. Vineet, P. Harish, S. Patidar, and P.J. Narayanan, “Fast Minimum Spanning Tree for Large Graphs on the GPU,” Proc. High Performance Graphics Conf. (HPG), pp. 167-171, 2010.

Y. Wang, G. Cong, G. Song, and K. Xie, “Community-Based Greedy Algorithm for Mining Top-K Influential Nodes in Mobile Social Networks,” Proc. ACM Int’l Conf. Knowledge Discovery and Data Mining (SIGKDD), pp. 1039-1048, 2010.

R. Zafarani and H. Liu, “Social Computing Data Repository at ASU,” http://socialcomputing.asu.edu/, Nov. 2012.

Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C., 2009. Introduction to Algorithms. 3rd ed. MIT Press.

Kleinberg, J. and Tardos, É., 2006. Algorithm Design. Pearso

IIT JAMMU

Dr. Suman Banerjee

Assistant Professor

Department of Computer Science and Engineering

Indian Institute of Technology, Jammu, India

10-Aug-2024

EXPERIENCE CERTIFICATE

This is to certify that, **Ms. Shruti Jain** student of **JK Lakshmipat University, Jaipur** was working as a research **intern** from 15th May 2024 to 27th July 2024 under the supervision of myself in online mode at Indian Institute of Technology Jammu. She worked on **"Maximizing Neutrailty in News Ordering"**. She has conducted herself well during the internship period. I wish her all the best for the future endeavours.

Suman Banerjee

Dr. Suman Banerjee,

Assistant Professor,

Department of Computer Sc. and Engineering,

Indian Institute of Technology, Jammu.